

Oncogenic Viruses

Louise S. Thai, MD • 36-slide current deck • viral oncogenesis → EBV → HPV
→ HHV-8 → other oncogenic viruses

4 official objectives

1 ScholarRx brick

EBV oncogenesis

HPV E6/E7

Kaposi histopathology

SOURCE PRIORITY: the current 36-slide Thai deck controls terminology, examples, mechanisms, histopathology, and faculty emphasis. The matching MCG 2029 AI Notes handout “Lecture 9: Oncogenic Viruses” is secondary enrichment. Board-style additions such as starry-sky histology, CD21, Tax, and some broader cancer associations are labeled Handout-only enrichment when they are not shown in the current deck.

Official Week 8 Objectives

1. Describe the general characteristics of oncogenic viruses.
2. Explain mechanisms of viral oncogenesis and relevant histopathology.
3. Discuss the spectrum of diseases caused by oncogenic viruses.
4. Explain treatment and prevention of cancer-causing viruses.

ASSIGNED SCHOLARRX

Oncogenic Microbes and Their Role in Cancer

Slides 3-7: How Can a Virus Cause Cancer?

CORE LANGUAGE

Proto-oncogenes are the gas pedal. Tumor suppressors are the brakes. Viral oncogenesis can increase growth signaling, activate oncogenes, integrate into host DNA, or interfere with tumor-suppressor pathways.

Regulation of Cell Cycle

- **Tumor suppressor genes** keep cell division in check: they're the "brakes" of the cell.
- **Proto-oncogenes** encode proteins that promote cell division: they're the "gas pedal" of the cell.
- Mutations $\Rightarrow$ uncontrolled cell proliferation.
- Tumor suppressors: **p53 & RB**
 - p53 protein allows the cell to repair the DNA or move it to apoptosis if there is a major damage.
 - p53 mutations are present in 1/3 of malignancies: bone and soft tissue sarcoma, colon cancer etc.
 - RB mutations lead to retinoblastoma etc.

Slide 5. The lecture's cell-cycle framing: proto-oncogenes promote division; tumor suppressors restrain it. p53 and RB are the two named tumor suppressors.

Cell-cycle control	Lecture concept
Proto-oncogene / oncogene	Growth-promoting signal; viral insertion or stimulation can drive excessive proliferation.
p53	Allows DNA repair or directs severely damaged cells toward apoptosis.
RB	Restrains cell-cycle progression; loss removes a major brake on division.

Major Oncogenic DNA Viruses

- Epstein-Barr Virus (EBV): *Burkitt & Hodgkin Lymphoma*
- Human Papilloma Virus (HPV): *Cervical cancer*
- Human Herpes Virus 8 (HHV-8): *Kaposi sarcoma*
 - *HHV-8 is aka KSHV (Kaposi sarcoma-associated herpes v.)*
- Hepatitis B Virus (HBV): *Hepatocellular Carcinoma*

Slide 6. Major oncogenic DNA viruses named in the deck: EBV, HPV, HHV-8/KSHV, and HBV.

Major Human Oncogenic RNA Viruses

- Retroviruses: After infecting the cell, the virus is reverse-transcribed into provirus DNA & incorporated into the host genome:
 - Human T-cell Leukemia Virus 1 (HTLV-1): *Lymphoblastic Leukemia*
 - HTLV – 2 : *T- cell lymphoma*
 - HTLV – 3 : *B-cell lymphoma*
- Hepatitis C Virus: *Hepatocellular Carcinoma*

Slide 7. Current-deck RNA-virus list includes retroviral and hepatitis-associated examples. Use the deck's wording for lecture-specific testing; the handout adds a more conventional board-style HTLV-1 framing.

Current-deck vs handout mismatch: slide 7 lists **HTLV-1: lymphoblastic leukemia, HTLV-2: T-cell lymphoma, HTLV-3: B-cell lymphoma**. The MCG 2029 handout instead emphasizes the board-style association **HTLV-1 → adult T-cell leukemia/lymphoma via Tax**. This companion does not silently merge those statements: the deck wording is faculty content; the Tax/ATLL connection is labeled handout enrichment.

Handout-only board framing: DNA tumor viruses commonly remove tumor-suppressor “brakes,” while retroviruses may activate host growth pathways through proviral insertion or viral regulatory proteins. The handout specifically reviews RB holding E2F at the G1/S checkpoint and calls p53 the “guardian of the genome.”

Slides 8-18: Epstein-Barr Virus (EBV)

EBV BACKBONE

Enveloped dsDNA herpesvirus → saliva → oropharynx → B-cell infection → latency in B lymphocytes.

Pathogenesis

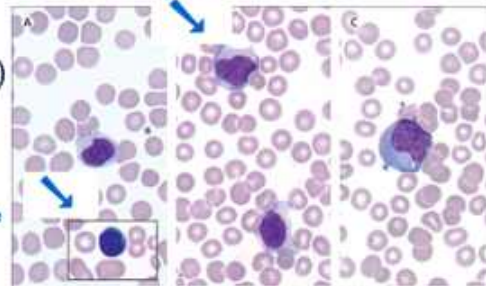
- Primary site of infection: Oropharynx (epithelial cells, lymphoid tissue)
- Viremia → Infection of B cells
- *Cytotoxic (CD8) T cells* are mobilized → *Atypical lymphocytes (Downey cells)* → infectious mononucleosis
- EBV infection in patients with T-cell immunodeficiency causes lymphoproliferative disease

Slide 9. EBV pathogenesis in the current deck: oropharyngeal infection, viremia, B-cell infection, CD8 T-cell response producing atypical/Downey cells, and increased risk of lymphoproliferative disease when T-cell immunity is impaired.

EBV Mononucleosis

- Age dependent: Adolescent or young adult
- Fever, fatigue, sore throat
- Exudative pharyngitis, lymphadenopathy, splenomegaly
- Lymphocytosis with **Atypical/Reactive Lymphocytes**:
- Elevated liver enzymes
- Heterophile antibodies (Mono-spot)

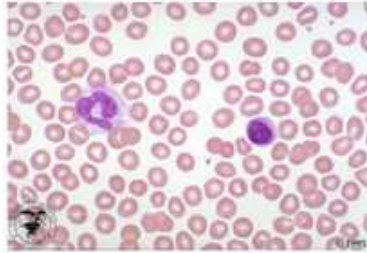
Normal Lymphocyte



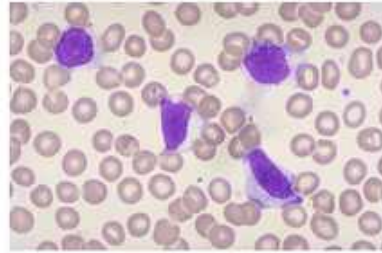
Slide 10. EBV infectious-mononucleosis pattern: adolescent/young adult, fever, fatigue, sore throat, exudative pharyngitis, lymphadenopathy, splenomegaly, atypical lymphocytosis, elevated liver enzymes, and heterophile antibodies.

Peripheral Blood Smear in Infectious Mononucleosis

Normal Lymphocyte



Atypical Lymphocytes



Slide 11. Peripheral blood smear comparison between a normal lymphocyte and atypical/reactive lymphocytes in infectious mononucleosis.

EBV Oncogenesis

- **LMP-1** (Latent membrane protein 1) as oncogene:
 - Acts as a constitutively active CD40 receptor, engaged in helper T-cell signals that stimulate **B-cell growth**
 - Prevents apoptosis by activating **BCL2** → **Immortalize** B cells
- **EBNA-2** (Epstein-Barr nuclear antigen):
 - Transactivates host genes, i.e., cyclin D → **Immortalize** B cells
- **vIL-10** (viral IL-10):
 - Prevents macrophages from activating T cells and is required for EBV-dependent transformation of B cells

Slide 13. EBV oncogenic proteins on the faculty slide: LMP-1, EBNA-2, and viral IL-10.

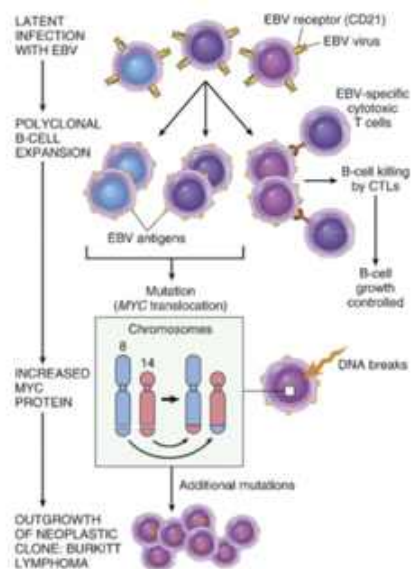
EBV factor	Current-deck function
LMP-1	Acts like a constitutively active CD40 receptor; stimulates B-cell growth; activates BCL2 and prevents apoptosis.
EBNA-2	Transactivates host genes such as cyclin D and promotes B-cell immortalization.
vIL-10	Reduces macrophage activation of T cells and supports EBV-dependent B-cell transformation.

Burkitt lymphoma

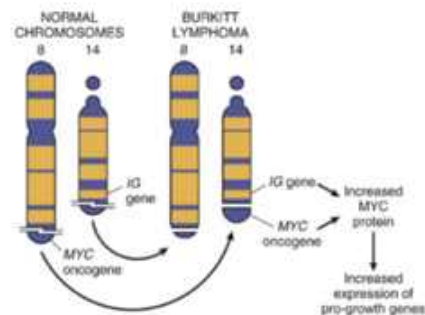
Burkitt Lymphoma

- Neoplasm of B lymphocytes (*B-cell lymphoma*)
- Prevalent along *malaria belt*
- The most common childhood tumor in central Africa and New Guinea
- More than 90% of African tumors carry the *EBV genome*
- Serum *antibody titers* against viral capsid antigens correlate with risk of developing Burkitt Lymphoma
- Chromosomal translocation of *myc oncogen*
- EBV is not directly oncogenic
- EBV acts as a polyclonal B-cell mitogen to set the stage for *8;14 translocation* and other mutations that ultimately produce cancer

Slide 14. Faculty nuance: endemic Burkitt is strongly EBV-associated, but the slide explicitly says EBV is not directly oncogenic here. EBV acts as a polyclonal B-cell mitogen that helps set the stage for MYC translocation and additional mutations.



Burkitt Lymphoma: HYI



Slide 15. The lecture's high-yield Burkitt figure centers on the MYC/Ig translocation, classically chromosomes 8 and 14, followed by increased MYC expression and neoplastic B-cell outgrowth.

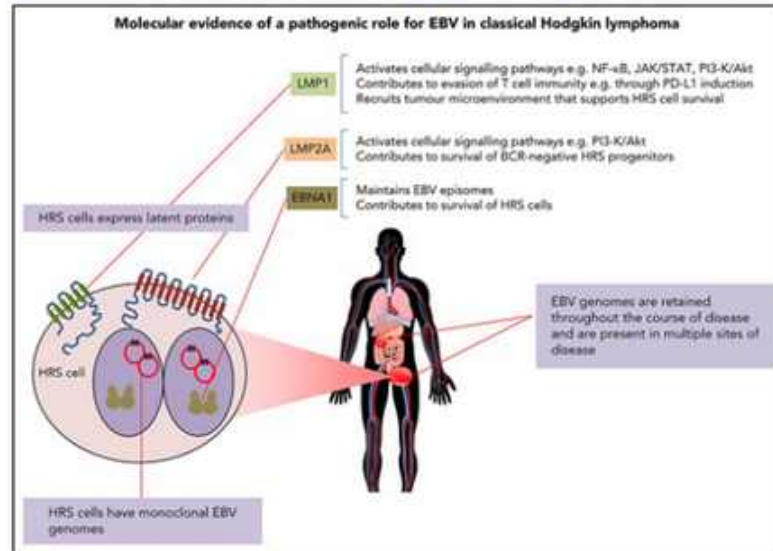
DO NOT FLATTEN THE EBV STORY

Slide 13: EBV has transforming latent proteins.

Slide 14: in Burkitt lymphoma specifically, the deck emphasizes that EBV is *not directly oncogenic*; it creates a proliferative setting in which **MYC t(8;14)** and other mutations produce cancer.

Handout-only enrichment: EBV binds **CD21** on B cells; endemic Burkitt often presents as an African child with a jaw mass and “starry-sky” histology. The handout also reviews Reed-Sternberg morphology and posterior cervical lymphadenopathy in EBV mono. Those are useful board links, but the current oncogenic-virus deck does not teach each of them as dedicated slides.

Hodgkin Reed Sternberg (HRS) Cells and EBV



Slide 17. The deck visually links EBV latent proteins to Hodgkin Reed-Sternberg cells and survival signaling.

EBV-associated disease on current deck	Faculty emphasis
Burkitt lymphoma	B-cell lymphoma, malaria-belt association, MYC t(8;14)
Hodgkin lymphoma	EBV association shown with HRS-cell latent proteins
Non-Hodgkin lymphoma	Included in EBV neoplasia spectrum
Nasopharyngeal carcinoma	Geographic association noted
Oral hairy leukoplakia	Shown as an EBV-associated condition in AIDS

Cross-lecture treatment conflict: this oncogenic-virus slide set lists EBV management as “symptomatic and antiviral (acyclovir).” The current Week 8 Herpesvirus lecture says *no specific antiviral therapy is beneficial* for EBV infectious

mononucleosis. The companion preserves the Lecture 5 slide wording but flags the mismatch rather than pretending the two current decks agree.

Slides 19-26: Human Papillomavirus (HPV)

Pathogenesis: *HYI*

- HPV infects squamous epithelial cells
- Hallmark: cytoplasmic vacuolization (koilocytosis)
- Benign warts: Viral DNA maintained in a nonintegrated episomal form.
- Oncogenesis in Cancers:
 - Viral DNA integrated in the host genome $\rightsquigarrow$ *malignant transformation*
 - Early genes, *E6 and E7*, are implicated
 - *E6 and E7* proteins inactivate tumor suppressor gene products – *P53 and RB*
 - Inactivation of P53 and RB leads to malignant transformation of the cells

Slide 20. The key mechanistic slide: HPV infects squamous epithelium; benign warts maintain viral DNA in an episomal form, whereas oncogenic transformation is associated with viral integration and expression of E6/E7.

HPV CANCER MECHANISM

E6 disables p53. E7 disables RB. Removing both brakes allows continued proliferation, survival of damaged cells, and genomic instability.

HPV-Disease Spectrum: HYI

- HPV 1-4: Skin and plantar warts
- *Low-Risk Types HPV 6,11 ➡ Anogenital Warts:*
 - *Condylomata acuminata* (cauliflower-like lesions)
 - Most common STD in the U.S. (> 42 million Americans)
 - 13 million new cases each year
 - Condoms do not prevent transmission
- *High-Risk Types HPV 16, 18, 31, 33 ➡ CANCER!*
 - Responsible for 70% of cases of *Cervical Carcinoma*
 - Anogenital, oral, and oropharyngeal carcinoma

Slide 21. Disease spectrum: HPV 1-4 skin/plantar warts; low-risk 6/11 anogenital warts; high-risk 16/18/31/33 cancer associations.

HPV 6, 11: Condylomata Acuminata (HYI)

Condylomata acuminata



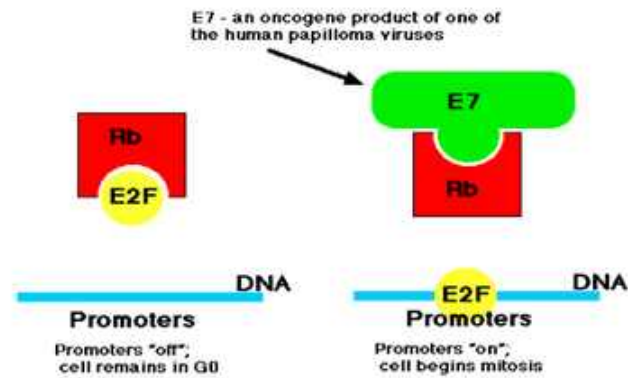
*Not to be confused with
condylomata lata:*



Figure 4: Condylomata lata in secondary syphilis (Supplied by DermNet NZ/Dr John Adams)

Slide 22. Faculty visual distinction: condylomata acuminata from HPV 6/11 should not be confused with condylomata lata.

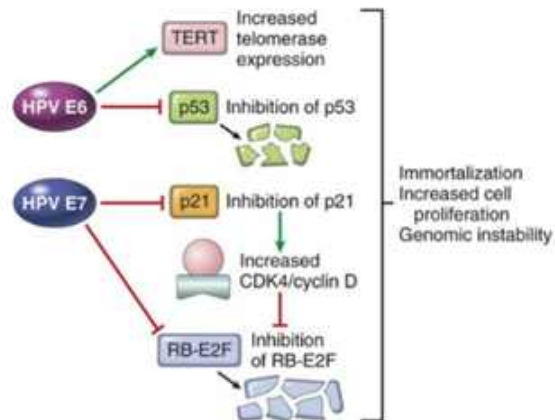
HPV Oncogenesis: *HYI*



Slide 23. E7 binds RB, releases E2F, and turns on transcription that promotes mitosis.

Ozzy checkpoint - click him once "E6 = p53 / E7 = RB" is automatic and you can explain why each one removes a different cell-cycle safeguard.

HPV E6 & E7 and Malignant transformation: *HYI*



Slide 24. The faculty E6/E7 summary broadens the mechanism: E6 increases telomerase and inhibits p53; E7 suppresses p21/RB-related control and increases proliferative signaling.

Diagnosis:

- Clinical
- Histopathology:
 - Koilocytosis also known as cytoplasmic vacuolization
 - Koilocytes are squamous epithelial cells with perinuclear halo
- DNA hybridization test



Slide 25. Histopathology: koilocytes are squamous epithelial cells with perinuclear halos/cytoplasmic vacuolization.

HPV group	Current-deck disease	High-yield feature
1-4	Skin and plantar warts	Benign wart types
6, 11	Condylomata acuminata	Low-risk anogenital wart types
16, 18, 31, 33	Cervical and other anogenital/oral/oropharyngeal cancers	High-risk types; 16/18 emphasized as major cervical-cancer drivers

Handout-only enrichment: the handout describes koilocytes as having a wrinkled “raisinoid” nucleus plus perinuclear halo and explicitly connects high-risk HPV to anal, vulvar, vaginal, and oropharyngeal carcinoma. Use this as board enrichment after the deck’s core E6/E7 + koilocytosis logic is solid.

Slides 27-32: HHV-8 / KSHV

Kaposi Sarcoma Herpes Virus (*KSHV*) aka Human Herpes Virus 8 (*HHV-8*)

- HHV-8 is an enveloped, double-stranded DNA virus
- Found in all types of Kaposi sarcoma
- Found in latent form within the *endothelial spindle cells* in the lesions
- Detected in saliva, semen, and peripheral blood mononuclear cells
- Transmitted sexually & via organ transplants
- Disproportionately affects HIV-infected MSM

Slide 27. HHV-8 is an enveloped dsDNA herpesvirus, found in all forms of Kaposi sarcoma. The deck places latent virus in endothelial spindle cells and names sexual and organ-transplant transmission.

HHV-8 CONCEPT

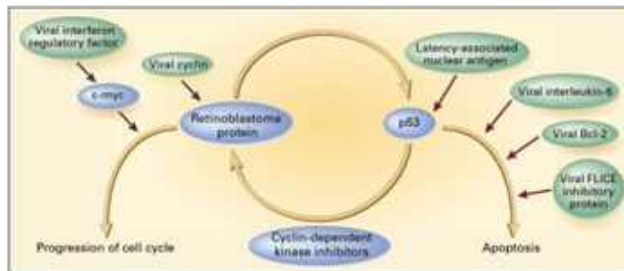
HHV-8 acts like a “molecular pirate”: it carries viral versions of growth/survival regulators that suppress apoptosis, interfere with tumor suppressors, and promote endothelial-cell proliferation.

KSHV/HHV-8 Oncogenesis:

- Molecular pirate – Has several human genes
- HHV-8 encodes anti-apoptosis proteins:
 - *BCL2* – Prevents apoptosis
 - Latency-associated nuclear antigen-1 (*LANA-1*) cyclin – Inhibits *RB* protein
 - vFLIP – FLICE-inhibitory proteins, inhibit apoptosis
 - Viral IL-6 – Blocks interferon signaling
 - IL-6, IL-8, chemokines – Growth factors for B lymphocytes and *endothelial spindle cells*
- Uncontrolled proliferation of *endothelial spindle cells* → *Kaposi sarcoma*

Slide 28. Faculty oncogenesis list: viral BCL2, LANA-1, vFLIP, viral IL-6, and additional cytokine/chemokine signaling drive survival and proliferation.

KSHV/HHV-8 Oncogenesis:



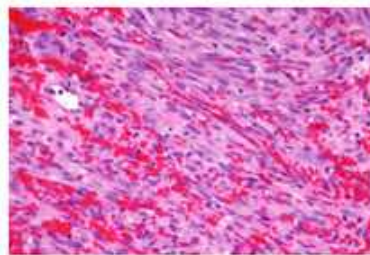
- HHV-8 Proteins interact with tumor-suppressors **RB** & **p53**.
- Viral interferon regulatory factor prevents interferon from repressing *c-myc*.
- FLICE = **F**as-associated death domain-**L**ike Interleukin-1 β -**C**onverting **E**nzyme
- **Red Arrows** indicate an inhibitory effect.

Slide 29. HHV-8 proteins interact with RB and p53, and viral interferon-regulatory signaling can keep c-MYC activity from being suppressed.

HHV-8 product / pathway	Current-deck function
Viral BCL2	Prevents apoptosis
LANA-1 / cyclin-related activity	Interferes with RB-linked cell-cycle control
vFLIP	FLICE-inhibitory anti-apoptotic activity
Viral IL-6 / IL-6 / IL-8 / chemokines	Growth signaling for B lymphocytes and endothelial spindle cells
Viral IFN regulatory factor	Prevents interferon-mediated repression of c-MYC

Kaposi Sarcoma

- Violaceous, raised plaques
- Histopathology:
 - Sheets of proliferating *endothelial spindle cells*, mostly in the dermis or subcutaneous tissues encompassing small vessels.
 - Slit-like spaces / Vascular channels containing RBCs → *H&E stain* →
 - Mitotic figures



Slide 31. Kaposi histopathology: violaceous raised plaques with proliferating endothelial spindle cells, slit-like vascular spaces/channels containing RBCs, and mitotic figures.

KAPOSI PATHOLOGY

Violaceous lesion + spindle endothelial proliferation + slit-like vascular spaces with extravasated RBCs = Kaposi sarcoma.

Handout-only enrichment: the MCG 2029 notes add primary effusion lymphoma and describe Kaposi as an endothelial malignancy that may involve skin, GI tract, and lung. The current oncogenic deck is centered on Kaposi sarcoma itself.

Slides 33-34: HBV and Retroviral Oncogenesis

Hepatocellular Carcinoma (HCC)

- HBV is responsible for 80% of all cases of HCC
- Latency period: 9 to 35 years
- Repeated cycles of cell injury and repair ⇔ Fibrosis ➔ HCC
- Integration of the HBV genome into the human chromosome may promote cell growth via:
 - Activation of oncogenes
 - Chromosome instability
 - Disruption of cell cycle regulator genes

Slide 33. HBV/HCC model in the current deck: repeated cycles of cell injury and repair drive fibrosis and eventual HCC; HBV genome integration may activate oncogenes, destabilize chromosomes, or disrupt cell-cycle regulators.

HEPATITIS-VIRUS CANCER LOGIC

Chronic injury → repeated regeneration → more opportunities for oncogenic mutation. The deck additionally gives HBV-specific integration mechanisms.

Retroviral Oncogenesis

- Insertion of the retroviral genome leads to conversion of *protooncogene* to *oncogene*
- Deletion
- Transduction
 - Exchange of reverse transcriptase gene for chromosomal protooncogene
 - The virus is no longer capable of replication
 - The virus is capable of transforming the cell
- Insertion of the virus into the host chromosome may promote transcription of adjacent oncogenes

Slide 34. Retroviral oncogenesis slide: proviral insertion can alter adjacent proto-oncogenes; the slide also describes deletion/transduction mechanisms and transformation after integration.

Handout-only board enrichment: the Drive handout emphasizes **HTLV-1 → adult T-cell leukemia/lymphoma via Tax** and frames both HBV and HCV hepatocarcinogenesis primarily through chronic inflammation/regeneration. These are kept separate from slide 7's current faculty list.

Treatment & Prevention

HPV

Current-deck approach	Details
Anogenital wart treatment	Imiquimod, podofilox, sinecatechins, cryotherapy, or surgical removal
Cervical-cancer screening	Pap smear
Vaccination	Gardasil-9; girls and boys may begin at age 9, routinely 11-12; protects against 6, 11, 16, 18, 31, 33, 45, 52, 58

Kaposi sarcoma

Treatment and Prevention

- Radiation and chemotherapy
- Antiretroviral therapy has greatly decreased frequency of KS in HIV infected patients
- Interferon- α and angiogenesis inhibitors like *vincristine*



Slide 32. The current slide lists radiation/chemotherapy, the major impact of antiretroviral therapy in HIV-associated disease, and interferon-alpha plus additional antitumor treatment approaches.

Source wording: slide 32 groups “interferon- α and angiogenesis inhibitors like vincristine.” This companion preserves the slide's treatment list but does not expand or reclassify those agents beyond what the source states.

EBV

- Current oncogenic-virus deck: peripheral smear/serology/Monospot/EBV-specific antibodies/PCR; treatment listed as symptomatic + antiviral (acyclovir); no vaccine.
- Remember the current Herpesvirus lecture gives a different EBV-mono treatment statement, as flagged above.

High-Yield Vignette Recognition

Vignette clue	Think	Mechanism / pathology
African child + B-cell tumor + MYC 8;14	EBV-associated Burkitt	EBV proliferative setting + MYC translocation
Cervical lesion + koilocytes	HPV	E6 $\rightarrow$ p53 loss; E7 $\rightarrow$ RB loss
Cauliflower anogenital wart	HPV 6/11	Condylomata acuminata
Violaceous plaques in immunocompromised patient	HHV-8 / Kaposi	Spindle endothelial cells + slit-like vascular spaces
Long-standing hepatitis + HCC	HBV/HCV-related oncogenesis	Chronic injury/regeneration; HBV may additionally integrate
Retroviral cancer question asking about host-gene activation	Insertional oncogenesis	Proviral integration alters nearby growth-control genes

Summary of HYI Points: Oncogenic Viruses

Virus	Diseases	Oncogenesis/Pathology
EBV	- Burkitt Lymphoma in Africa - Hodgkin lymphoma - Non-Hodgkin lymphoma	MYC translocation 8;14
HPV 6,11	Anogenital Warts: Condylomata acuminata	Cervical CA: - E6 and E7 inactivate P53 and RB - Path hallmark: Koilocytosis
HPV 16, 18	Cervical carcinoma	
HHV-8	- Kaposi sarcoma in immunocompromised patients - AIDS patients	- RB and P53 suppression - FLICE inhibition - Path hallmark: Proliferation of endothelial spindle cells

Slide 35. Faculty final summary condenses the lecture to EBV/MYC, HPV 6/11 vs 16/18 with E6/E7 and koilocytosis, and HHV-8 with RB/p53 suppression and endothelial spindle-cell proliferation.

LAST 60 SECONDS

EBV: LMP-1 / EBNA-2 / vIL-10; Burkitt t(8;14).

HPV: E6 → p53, E7 → RB; koilocytosis; 6/11 warts vs high-risk types cancer.

HHV-8: anti-apoptotic/growth signaling; Kaposi spindle cells + slit-like vessels.

HBV: chronic injury + integration-related genomic disruption.

Source hierarchy: Thai 36-slide current deck is primary. Official Week 8 LOs supply the ScholarRx assignment. The MCG 2029 AI Notes “Lecture 9: Oncogenic Viruses” handout is secondary enrichment and is explicitly labeled when it goes beyond or differs from the current deck.